ORIGINAL MEMOIR · BOVINE TUBERCULOSIS · TRANSLATED FROM THE FRENCH

VACCINATION OF CATTLE AGAINST TUBERCULOSIS

and a new method of prophylaxis for bovine tuberculosis

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EDITORIAL NOTE

Complete English translation of the original French memoir, reset for reading and reference. Nomenclature, dosages, dates and the authors' own emphases follow the source; footnotes are given where they occur. The six temperature plates of the original are clipped from the journal and reproduced at the points where the authors cite them.

SOURCE

Vaccination des bovidés contre la tuberculose et méthode nouvelle de prophylaxie de la tuberculose bovine. Annales de l'Institut Pasteur, 38e année, n° 5, mai 1924, pp. 371–398.

In a series of memoirs published successively over the past twenty years,[1] we have established a certain number of facts which allow us to understand better than was hitherto possible the mechanism of bacillary infection, and the conditions to be fulfilled in order to confer upon susceptible organisms resistance to contamination, whether natural or artificially provoked.

Let us recall some of the most important of them.

In August 1906 and July 1907, in these *Annales*, we showed that multiple reinfections, or massive single infections, surely determine the evolution of a fatal tuberculosis in cattle, whereas the disease produced by a single contamination usually heals if the animal is kept sheltered from accidental contagion.

From that time onward our researches supplied the proof that animals thus healed, to the point of no longer reacting to tuberculin, are no longer susceptible — at least for a certain time — of being reinfected, even with doses of virulent bacilli capable of giving the controls an acute granular tuberculosis, fatal within a few weeks.

[1] These *Annales*, 1905, 1906, 1907, 1908, 1913, 1914, 1920.

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Original French text published in the Annales de l'Institut Pasteur; page numbers of the original are marked in the text.

Thus, we wrote, "we are naturally led to direct our researches towards obtaining vaccinal immunity against tuberculosis by introducing into the lymphatic system of the organism, by the normal routes of infection, that is to say by the digestive tract, tuberculous bacilli attenuated, modified, or deprived of virulence."[1]

At the same time H. Vallée, studying with H. Rossignol[2] the effects of the antituberculous vaccination method that Behring had proposed, observed in one experiment that "a first benign and healed attack of tuberculosis confers upon the organism a marked, though incomplete, resistance to a severe reinfection."

Römer's experiments,[3] later than ours and carried out on sheep and guinea pigs, arrived at the same conclusions.

Later[4] we showed that the tolerance of the organism towards a tuberculous infection or towards reinfections is a function of the presence, in that organism, of living bacilli, virulent or not; and that this tolerance can be conferred artificially upon young cattle by introducing into their blood circulation a suitable dose of our bovine bacillus, called the "bile bacillus", non-virulent and non-tuberculigenic, but still endowed with a certain toxicity and capable of provoking the formation of antibodies.

This "bile bacillus" is well enough known that there is no need to dwell upon its acquired characters. Let us recall only that it has for its origin a highly virulent bovine strain, called "Nocard's milk", transplanted for the first time onto potato cooked in ox bile (glycerinated at 5 per 100) in January 1908, and reseeded since upon that same medium about every fifteen days.[5]

After four years our bacillus was no longer virulent for the ox. It still was for the horse.

After thirteen years of successive cultures upon biliary potatoes (5 January 1921: 230th passage), it was completely deprived of virulence, even at strong doses, for all animal species. *It had become incapable of provoking the formation of tubercles by intravenous, intraperitoneal or subcutaneous inoculation, and equally by ingestion.*

At that last date, in order to fix its precious qualities definitively, we transferred it and henceforth preserved it upon ordinary glycerinated potato.

[1] These *Annales*, August 1906, p. 623.

[2] *Bull. de la Soc. de Méd. vétér. prat.*, report on the Melun experiments, 14 March 1906.

[3] *Sitzungsber. d. ärztl. Vereins zu Marburg*, 19 May and 22 July 1908; 21 May 1909.

[4] These *Annales*, February 1913, April 1914, September 1920.

[5] The aim we pursued in cultivating the tubercle bacillus without interruption in the presence of ox bile was to modify hereditarily the physico-chemical constitution of the bacillus, by inducing it to develop in an extremely alkaline medium particularly rich in lipoids (cholesterol 0.410 to 0.813 per 1,000; lecithins and neutral soaps 0.690 to 1.317 per 1,000). We hoped thereby to render the bacillary envelopes and protoplasm more easily digestible by the phagocytes, or at least more tolerable to them. The facts have shown that we were on the right road.

From the first culture, which was abundant, we saw reappear the aspect and the odour known of tuberculosis cultures, although to the touch of the spatula the bacilli are less dry, more unctuous, easier to spread. Immediately too, the culture was accompanied by the production of tuberculin, a quality it had seemed to lose in media in the presence of bile. We made a glycerinated decoction of this tuberculin: it behaves, towards tuberculous cattle, like the best tuberculin obtained from virulent bacilli.

Every five passages upon ordinary glycerinated potato, the culture was inoculated at doses of 1 to 10 milligrammes under the skin of guinea pigs. Doses above 3 milligrammes determined a small abscess, healing spontaneously, which was accompanied by no engorgement of the corresponding inguinal node. None of our animals became tuberculous, and *we have never succeeded, by successive re-inoculations, in restoring tuberculigenic properties to our bacillus.*

We are therefore in possession of a truly avirulent race of bovine bacillus. This bacillus, although toxic for the tuberculous animal and a producer of active tuberculin, has lost all aptitude for creating tubercles. It is completely harmless for all animal species, including the chimpanzee, and also for man, even by intravenous injection at the dose of forty million living microbial bodies (1 milligramme).

It is with this bacillus, designated henceforth, for convenience of language, by the initials **BCG**, that we carried out the experiments which follow and the preventive inoculations reported further on.

I INTERPRETATION OF TUBERCULIN REACTIONS IN VACCINATED ANIMALS

On many occasions in the course of our previous researches, and notably in those related in our memoir of September 1920,[1] we had been struck by the rapidity with which our animals vaccinated by the intravenous route ceased to react to tuberculin, even while they remained perfectly tolerant (up to a maximum of eighteen months) towards virulent bacilli likewise introduced by the intravenous route, or absorbed through close and continuous cohabitation with tuberculous cattle. We thought at the time that this tolerance must be linked to the presence of the avirulent bacilli in the organism, and that it ceased when the elimination of those bacilli by the normal routes of excretion (hepatic gland, intestine and mammary glands) was complete.

If this were so, one ought to try to prolong that period of vaccinal impregnation by introducing the avirulent bacilli no longer into the veins but *under the skin* — a procedure moreover simpler and more practical, and less uncertain than absorption by the digestive routes — and by renewing the subcutaneous injection every year, for example, since it appears to be harmless.

This is what we attempted in the following experiment.

[1] These *Annales*, 1920, p. 553.

EXPERIMENT. — Sixteen Breton heifers aged seven to eight months, free of tuberculosis, are divided into two lots: eight of them receive the vaccinal inoculation into the veins, the eight others receive it under the skin of the dewlap. Tuberculin tests are carried out, for all the animals of each lot, two months and six months after the inoculation.

TABLE I — TUBERCULIN REACTIVITY BY ROUTE OF VACCINAL INOCULATION

TUBERCULINATION	VEINS · AFTER 2 MONTHS	VEINS · AFTER 6 MONTHS	DEWLAP · AFTER 2 MONTHS	DEWLAP · AFTER 6 MONTHS
Positive reaction	3	0	8	5
Doubtful reaction	1	0	0	3
Negative reaction	4	8	0	0

The observation is clear-cut: animals vaccinated by inoculation under the skin retain the faculty of reacting positively to tuberculin longer than animals vaccinated in the veins.

We have multiple reasons for admitting that this persistence of sensitivity to tuberculin is correlative of the symbiotic life of the tubercle bacillus with the cellular elements — not the mobile ones, but the fixed ones.

From this symbiotic life there results, as one of us has written elsewhere,[1] a cellular complex analogous to a lichen, which is the product of the symbiosis of an alga and a fungus. When this complex is realised, and so long as it subsists, so long as it escapes the defensive phagocytic actions, the organism of which it is the parasite reacts in a characteristic manner to fresh contributions of tuberculin or of bacilli: this is what we call the "phenomenon of Koch", because its observation led Robert Koch to the discovery of tuberculin.

It appears therefore that resistance to tuberculous infections or reinfections — resistance natural or artificially provoked — must translate itself, like the infection itself, by sensitivity to tuberculin.

Consequently, the positive tuberculin reaction can no longer be considered as a criterion of infection. It is rather, in cattle as in man, a criterion of immunity. It reveals only the existence, somewhere in the organism which reacts, of an active or latent focus, recent or old, of symbiotic life of a bacillus, virulent or non-virulent, with some lymphatic cell, macrophage or giant cell.

By that fact it definitively loses, if not all interest, at least all diagnostic importance, since on the day when all subjects — men or animals susceptible to tuberculosis — are, with a prophylactic aim, artificially impregnated with avirulent bacilli, they will react in the same manner as infected subjects.

[1] A. Calmette, *L'infection bacillaire et la tuberculose*, Masson, 2nd edition, Paris, p. 108.

II FATE OF THE VACCINE BACILLI INTRODUCED UNDER THE SKIN

If one injects into cattle free of tuberculosis, under the skin of the dewlap, a weak dose of vaccine bacilli, such as 2.5 milligrammes, no local reaction is observed around the inoculated point, and at no moment thereafter does the animal react to the tuberculin test. This dose is insufficient to confer upon it any appreciable resistance to a later infection by an intravenous injection of virulent bacilli.

If, on the contrary, the quantity of vaccine bacilli injected is more considerable, from 50 to 100 milligrammes, a local lesion is produced almost immediately, upon whose evolution and persistence we shall return further on in relating our vaccination experiments.

This lesion has no tendency whatever to suppurate, and it most often ends by disappearing without leaving traces.

One heifer, however, which had received 100 milligrammes of vaccine bacilli in the dewlap and which ten months later furnished a positive intradermal reaction, retained around the inoculated point a lesion of cystic appearance, as large as a pigeon's egg, slightly fluctuating and floating freely in the connective tissue. A puncture with the syringe, after shaving the skin, permitted the withdrawal of about 1.5 cubic centimetres of serous, faintly opalescent liquid, containing no leucocyte but rich in granular bacilli imperfectly stainable by Ziehl. One part of this liquid served to seed three tubes of Petroff's medium; the other was diluted in 2 cubic centimetres of sterile water and inoculated under the skin of the thigh of four guinea pigs. None of the tubes gave a culture, and the four guinea pigs remained free of any local or ganglionic lesion; sacrificed three months later, they were found perfectly healthy.

One cannot conclude from this that the vaccine bacilli were dead; but they were at the very least degraded enough to have lost their vitality.

Let us retain for the moment that *this massive inoculation of vaccine bacilli is immediately, and remains subsequently, harmless.*

It seemed to us of interest to investigate whether virulent tubercle bacilli, placed in the same conditions — that is to say withdrawn from the direct action of the leucocytes — underwent the same degradation.

We know already that such bacilli, enclosed in collodion sacs whose impermeability must be absolute, and placed in the peritoneum of rabbits, not only do not multiply but cease after some months to be inoculable. To realise the experiment in the ox, we charged, under vacuum, balls of pumice stone of 10 millimetres in diameter with a concentrated emulsion of virulent bovine bacilli. The charge retained by one ball was 2.5 milligrammes of bacilli.

After incision of the skin of the dewlap of an ox and lacerating of the subcutaneous connective tissue with the cannula of a large trocar, a ball thus charged was pushed deeply under the skin through the lumen of the cannula. The operative wound, sealed with Michel clips, healed by first intention. The mobile foreign body was easily perceived in the connective tissue, and around it a weak cellular reaction.

At no moment — that is to say after one, three and six months — did the animal react to tuberculin.

Six months after the insertion of the ball, it was removed very easily by a simple incision at its level. It was retained only by a few weak connective adherences. A continuous capsule, barely 1 millimetre thick, surrounded it, exactly applied to its surface, without interposition of liquid. Scraping of the inner face of this capsule showed no leucocyte. The ball was crushed and triturated in an agate mortar in 5 cubic centimetres of physiological water. The product of this trituration showed under the microscope a considerable number of bacilli, many of them granular but easy to stain by Ziehl. Injected under the skin of four guinea pigs, it left these animals free of infection.

As for the ox bearing this ball, it was inoculated, on the very day of the removal, with 5 milligrammes of virulent bacilli by the intravenous route. It succumbed fifty-eight days later to an acute pulmonary granulie, having consequently acquired no resistance.

It is therefore because, in the conditions of this experiment, no symbiosis had been established between the bacilli and the cells, so that the "phenomenon of Koch", translating resistance to reinfections, could not manifest itself.

We have moreover been able to establish that, even in the guinea pig, the demonstration of this "phenomenon" is correlative of the dose of reinfecting bacilli.

> EXPERIMENT. — Four guinea pigs receive under the skin of the right thigh 0.1 milligramme of virulent bacilli (weighed fresh). Eighteen days later, while the local lesion is evolving, they receive, at the same time as four controls, under the skin of the left thigh, 0.02 milligramme of the same bacilli. Thirty days after this inoculation, while the controls all present the characteristic inguinal node, hard and mobile, none of the four guinea pigs reinfected with this weak dose has presented a necrotic local lesion. The skin around the inoculated point remains supple, without trace of underlying infiltration, and the inguinal nodes are imperceptible.

In such conditions the reinfecting bacilli are therefore very well tolerated.

If, on the contrary, the dose of reinjected bacilli is appreciably equal to or greater than that which realised the first infection, the tuberculous organism does not support them. It rids itself of them by the rapid formation of an abscess which soon empties to the exterior, then cicatrises. This is then the typical phenomenon of Koch, such as one observes it in cattle reacting to tuberculin into which one injects, under the skin, a strong dose of bacilli even slightly virulent.

It doubtless happens often that cattle reacting to tuberculin, spontaneously contaminated, offer a considerable resistance to reinfections, even massive ones. We find a remarkable example of this in the following experiment.

> EXPERIMENT. — In 1922 we possessed a four-year-old heifer in excellent condition, which had reacted for two years to the tuberculin test renewed every six months, and which we had kept throughout that long period completely isolated from any contamination of the neighbourhood. Spontaneous cure having therefore not occurred, we inoculate into the jugular vein 5 milligrammes of virulent bovine bacilli, a dose which constantly produces a fatal acute tuberculosis within a maximum of sixty days. From the ninth hour after inoculation the temperature rises to 41.5° and the animal is seized with trembling and violent fits of coughing. The fever remains high for six days, but diminishes little by little, and all returns to order. Appetite and condition are maintained. Four months later the heifer is slaughtered. At autopsy one finds seven tuberculous nodules, the size of millet grains, in a tracheo-bronchial node, and one caseo-calcareous tubercle the size of a hempseed at the anterior extremity of the right lung. The organs of the splanchnic cavity are free of lesions. With sterile instruments one takes one of the mesojejunal nodes, one of the nodes bordering the colic artery, the node of the ileo-caecal valve and one hepatic node. All these nodes, of normal appearance and volume, are triturated separately, and their expressed juice injected under the skin of sixteen guinea pigs (four for each node). One month later, all these guinea pigs without exception are bearers of the specific adenitis, and half of them have suppurated inoculation lesions.

It is evident that in our heifer it was the small tubercles of her tracheo-bronchial node and of her right lung which sensitised her to tuberculin, and it is highly probable that it is the reinfecting bacilli, so easily tolerated by her, that we recovered in the apparently healthy nodes of her splanchnic cavity. But what we wish above all to retain from this observation is that our heifer, afflicted with benign lesions of tuberculosis for more than two years, showed herself capable of supporting, without her disease being aggravated, an artificial reinfection by more than two hundred million bacilli of great virulence.

III VACCINATION EXPERIMENTS IN CATTLE BY MASSIVE SUBCUTANEOUS INOCULATION OF BCG

The critical examination of the foregoing facts, as well as the results of the multiple trials we have carried out — of which it would take too long to relate all the details — having shown us the manifest advantages of vaccination by the subcutaneous route, in one operation, with a massive dose of our avirulent, non-tuberculigenic bacillus (BCG), we undertook the following series of experiments.

> PROTOCOL. — Twenty Breton heifers of seven to eight months, free of tuberculosis, are divided into two lots. Twelve are vaccinated; eight, maintained in the same conditions, serve as controls. The twelve heifers of the first lot, numbered 1 to 12, are divided into six groups of two animals (1–2, 3–4, 5–6, and so on). Each group occupies a box isolated from the next. The odd numbers receive, into the loose connective tissue of the dewlap, a single inoculation of 100 milligrammes of BCG emulsified in 10 cubic centimetres of physiological water. The even numbers are inoculated in the same manner, but with 50 milligrammes of BCG in 10 cubic centimetres of physiological water. We decide that one control shall be assigned to each of these six groups, and that the virulent test inoculation shall be made to the animals of each group respectively one, three, six, twelve, fifteen and eighteen months after vaccination, at the same time as to its control.

A. REACTION OF THE ANIMALS TO THE VACCINAL INOCULATION

It was the same with 50 as with 100 milligrammes.

a) Local reaction. — Twenty-four hours after the inoculation one observes, in the connective tissue, the existence of an oedematous, soft mass, as large as a walnut or a hen's egg. On the following days this mass becomes denser, hard, smooth, mobile; in a few animals (one in twelve) an oedema of new formation reappears from the third to the eighteenth day and seems to envelop the little tumour; then all returns to order. This lesion has no tendency to suppurate.

One of our vaccinated animals (no. 10) having died twenty-five days after the inoculation, of ulcerative gastritis, permitted us to study the structure of the tumour at that moment. It was a hard mass, like rubber, 5 centimetres long and 2 centimetres thick, formed of fibrous, lardaceous tissue imprisoning islets of necrosis of a golden-yellow colour, resembling, so as to be mistaken for them, old tuberculous lesions of a node. Each of these islets is surrounded by a resistant capsule whose content, which enucleates easily, holds a considerable number of bacillary bodies, rather poorly stainable.

The vaccinal lesion thus constituted under the skin diminishes slowly in volume. It remains perceptible in the sixth month and tends to disappear from the tenth to the eighteenth. It is always painless.

b) General reaction. — The majority of our animals (nine out of twelve) presented no elevation of temperature, no disturbance of health or of appetite. In 25 per 100 of them (three out of twelve) we observed, from the fifteenth to the eighteenth day, a rather strong febrile rise (from 38.6° to 40.5°) accompanied by dullness and anorexia. The fever lasts five or six days, then disappears, and the general state becomes excellent again (fig. 1). This late hyperthermia recalls exactly what we observed in our trials of vaccination by the intravenous route.[1] It results without doubt from a bacillary septicaemia consequent upon the passage of a more or less large number of microbial bodies into the blood circulation.

c) Tuberculin reaction. — We preferred not to submit our vaccinated animals to periodic tuberculin tests by the subcutaneous route, in order not to modify their state of resistance or of sensitisation by an untimely habituation. The intradermal reaction not having the same drawback, we practised it exclusively.

This reaction remains positive so long as the vaccinal lesion of the connective tissue persists, at least so long as the latter remains clearly perceptible to the touch. It became negative only from the sixth month onward in two animals out of seven (29 per 100). At the twelfth month, a single one still reacted positively.

[1] These *Annales*, September 1920.

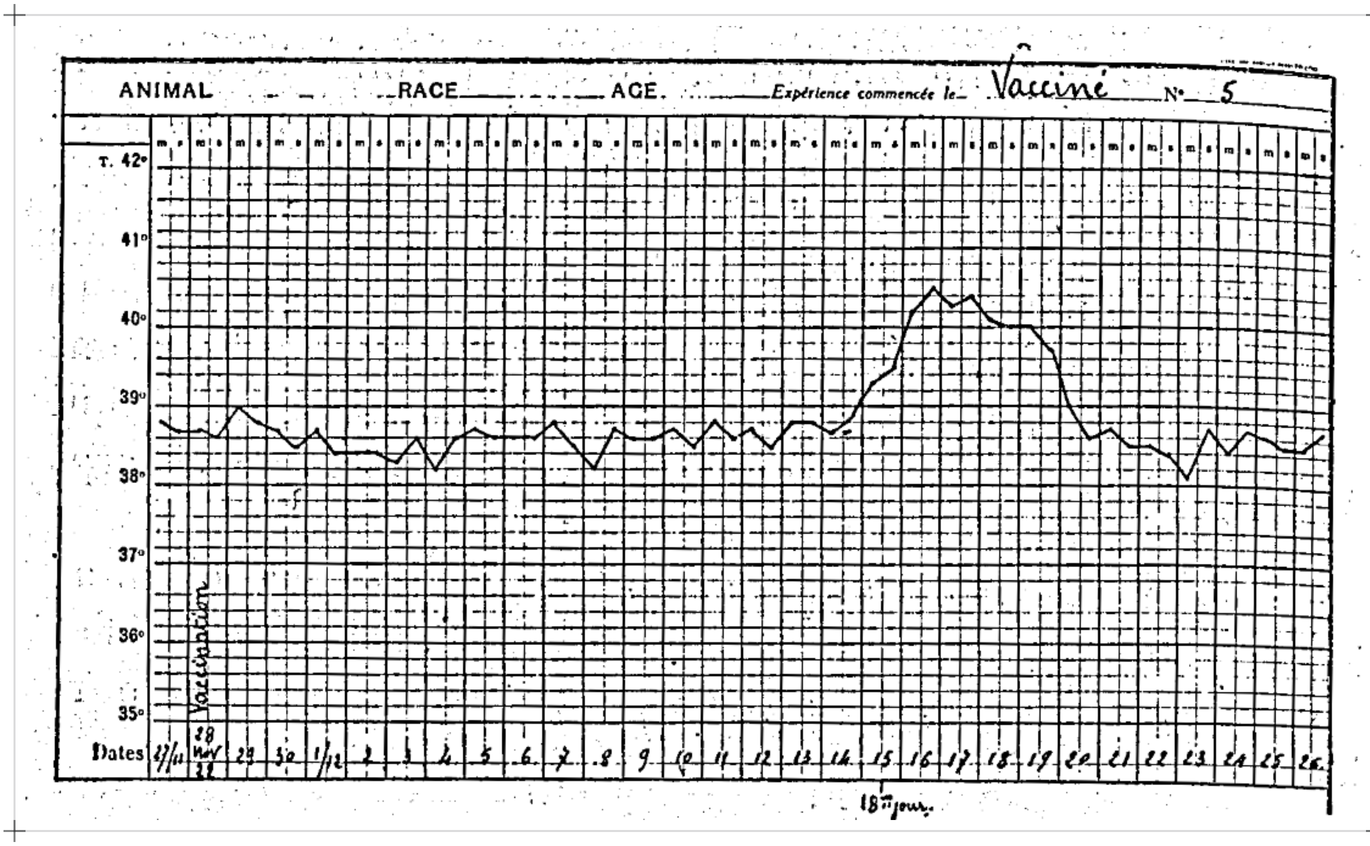

FIG. 1 — Vaccinated no. 5: the late febrile rise, fifteenth to eighteenth day after vaccination.

B. REACTION OF THE CONTROLS TO THE TEST INOCULATION

We used for our test inoculations a bovine bacillus isolated directly by one of us, from a pleural lesion at the abattoir, upon Petroff's medium, and of which 1/250 of a milligramme kills the guinea pig within an average interval of ninety days after injection under the skin.

The dose employed for the cattle was uniformly 5 milligrammes (weighed fresh) inoculated into the jugular vein.

In our control cattle the test inoculation was never followed by an immediate thermal reaction. It is only after an incubation period of thirteen to fifteen days that one sees a violent hyperthermia (2°) manifest itself abruptly, at the same time as the general state becomes less good. The temperature remains high, the cough is frequent, the number of respiratory movements increases to eighty and more. Death occurs from the thirtieth to the forty-fifth day (fig. 2). It is determined by a massive pulmonary granulie.

A few controls, by exception, showed a more precocious elevation of temperature, from the fourth to the sixth day (40°), lasting at most three days; then all returned to order until the fourteenth or fifteenth day, after which the ordinary thermal ascent occurred (fig. 3). But in these animals the clinical picture changes. After a

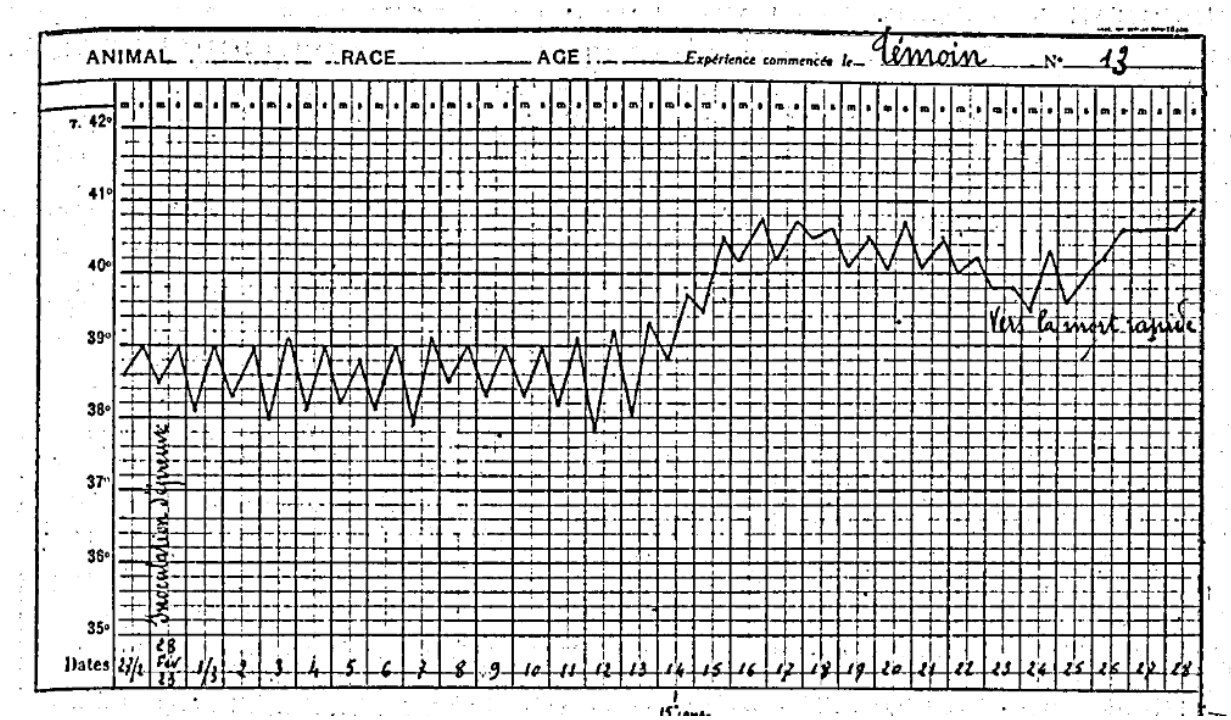

FIG. 2 — Control no. 13: incubation of thirteen to fifteen days, then the fatal thermal ascent.

hyperthermia lasting eight to twelve days, with a daily tendency to fall, the general state seems to improve; respiration is easier, the temperature becomes almost normal again. The fits of coughing persist, however, and the lesions of miliary tuberculosis evolve, provoking death only after several months. They are nevertheless always largely constituted by the end of the second month (fig. 4).

The early thermal reaction thus observed in a few fresh animals after the virulent intravenous inoculation appears to us to translate a particular state of resistance whose cause escapes us. Perhaps it is due to a very slight pre-existing bacillary infection which tuberculin had not revealed.

C. REACTION OF THE VACCINATED ANIMALS TO THE TEST INOCULATION

In vaccinated cattle the test inoculation is always followed immediately by a violent thermal reaction (40°–41°), whose maximum is observed from nine to twelve hours after the intravenous injection of the virulent bacilli (fig. 4).

Sometimes this reaction disappears very quickly, or else it persists for three to six days, fading gradually (fig. 5); most often, at the end of twenty-four hours everything has returned to order, and at no moment thereafter will the health of the animals be disturbed. Nothing will allow one to suspect the extremely severe test they have undergone.

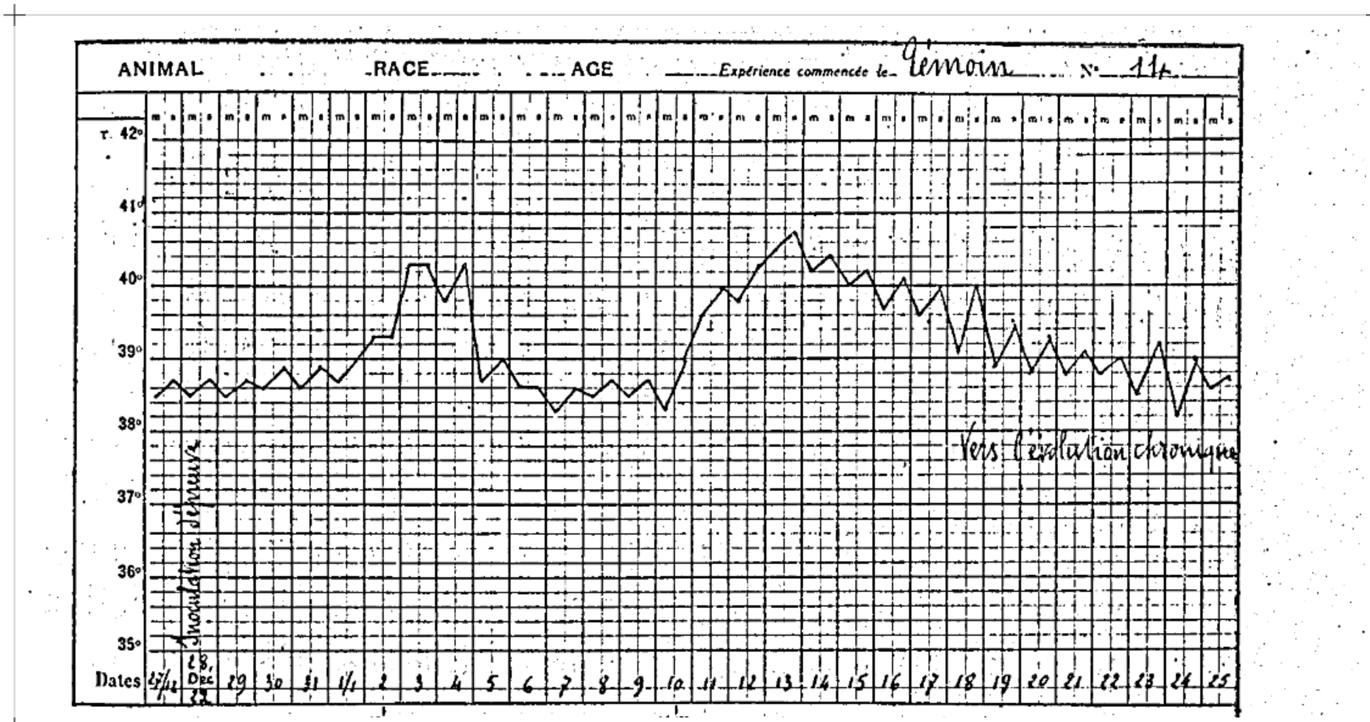

FIG. 3 — Control no. 14: an early febrile reaction on the fourth to sixth day, then a prolonged course.

This brutal, immediate reaction is constant, even in vaccinated animals which, a few days before the virulent test, have a negative intradermal reaction. It is certainly of tuberculinic nature, and here is the proof.

> Thirty-six hours before the virulent inoculation, in a vaccinated animal, we inject under the skin 3 cubic centimetres of diluted tuberculin. The thermal reaction develops from the tenth to the fifteenth hour after the injection. Twenty-four hours later we make the test inoculation with the virulent bacilli. This one is followed by no hyperthermia whatever, as figure 6 shows. It is exactly what one observes in tuberculous cattle which react strongly to a first tuberculination and no longer react if the same dose of tuberculin is reinjected two or three days afterwards.

D. RESULTS OF THE EXPERIMENTS FOR EACH GROUP OF VACCINATED CATTLE

GROUP I — VACCINATED NOS. 1 AND 2 · CONTROL NO. 13

The two vaccinated animals are tested at the same time as the control, *one month after vaccination*.

Control 13 is sacrificed, greatly emaciated, sixty days after the test. Its autopsy shows a miliary tuberculosis of the lungs with tubercles at all stages, from the pin's head to the caseous hempseed.

Vaccinated no. 2 was tested although it had, from time to time, fits of coughing resulting from its being afflicted with verminous bronchitis; it emitted a muco-purulent discharge containing embryos of strongyles. We tried without success to treat it by intratracheal phenicated injections. The animal became more and more cachectic and succumbed five months and seven days after the test inoculation. Its autopsy, apart from

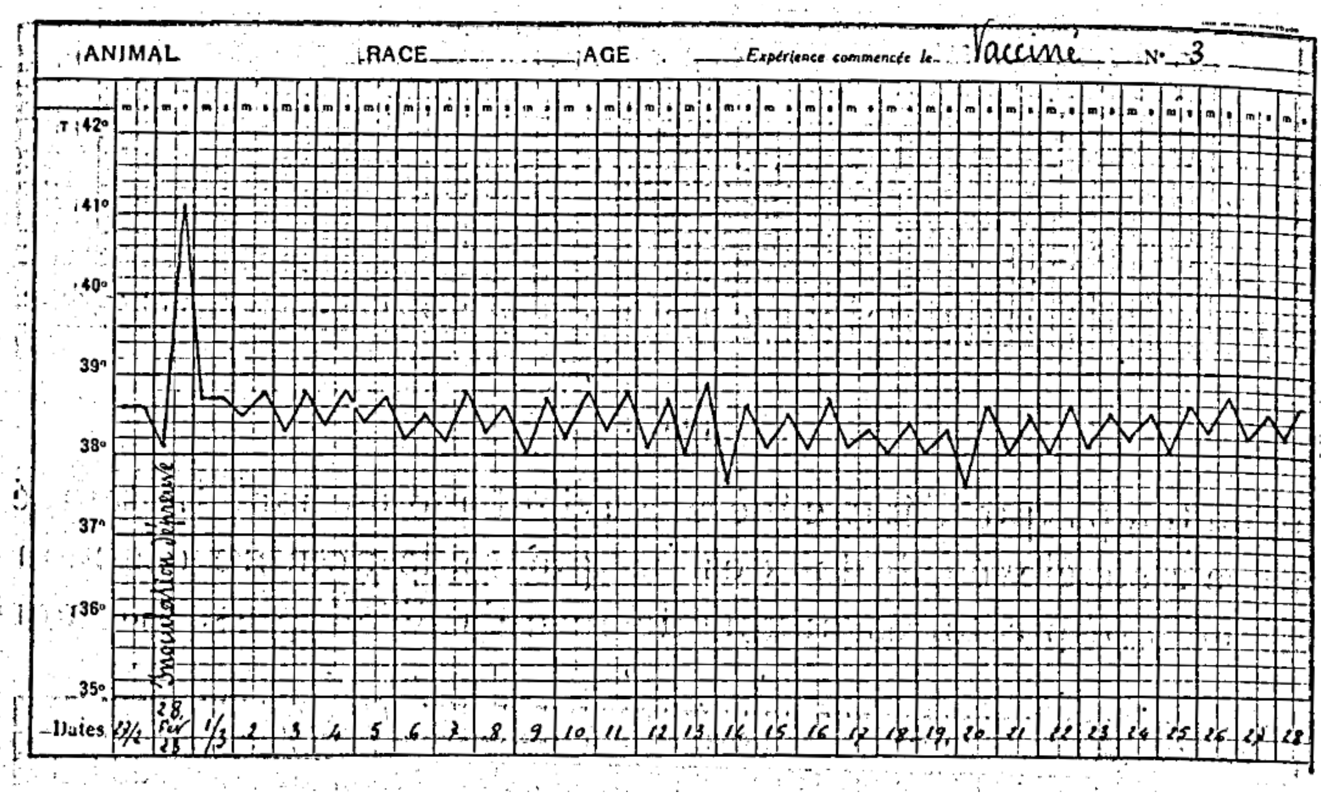

FIG. 4 — Vaccinated no. 3: the immediate reaction, nine to twelve hours after the virulent challenge.

the disorders caused by the parasites, did not permit the detection of the least tuberculous lesion. The left pretracheo-bronchial node is taken, triturated, expressed, and the product inoculated under the skin of four guinea pigs. Thirty days later, the four guinea pigs are bearers of the specific adenitis. This observation draws a particular interest from the fact that it concerns an animal in which, despite its complete state of decline ending in death by cachexia, the vaccinal protection all the same prevented the evolution of virulent tuberculosis.

Vaccinated no. 1, remaining in perfect condition, is slaughtered twelve months after the test. It is completely free of tuberculosis. The left pretracheo-bronchial node is taken, triturated, expressed, and the product inoculated under the skin of four guinea pigs. Forty-five days later, the four guinea pigs, which present no adenitis, are sacrificed. The most minute examination reveals in them no tuberculous lesion. We shall return presently to this fact, to which we attach great importance.

GROUP II — VACCINATED NOS. 3 AND 4 · CONTROL NO. 14

The two vaccinated animals are tested, at the same time as the control, *three months after vaccination*. Control no. 14 dies forty-four days after the inoculation: massive pulmonary granulie. Vaccinated 3 and 4, in

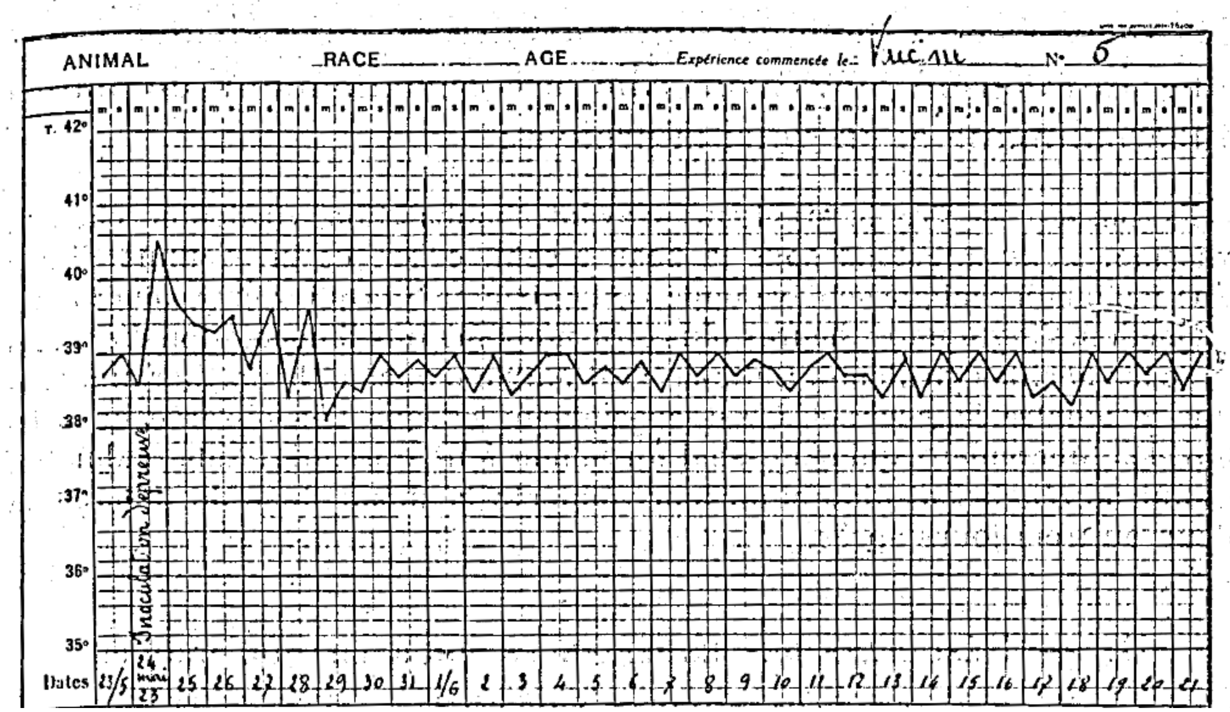

FIG. 5 — Vaccinated no. 6: the immediate reaction persisting several days before fading.

excellent condition, are slaughtered eleven months after the test. They are free of tuberculosis. Their left pretracheo-bronchial nodes are taken, triturated, expressed, and the products inoculated separately under the skin of four guinea pigs. Forty-five days later, the four guinea pigs inoculated with the node of vaccinated no. 3 are sacrificed and found free of infection. Those inoculated with the node of vaccinated no. 4, on the contrary, are bearers of the specific adenitis.

GROUP III — VACCINATED NOS. 5 AND 6 · CONTROL NO. 15

The two vaccinated animals are tested, at the same time as the control, *six months after vaccination*. Control 15, in very bad condition and incapable of standing, is sacrificed sixty days after the test inoculation: miliary tuberculosis of both lungs with foci of hepatisation. Vaccinated 5 and 6, in excellent condition, are slaughtered eight months after the test. They are found free of tuberculosis. Their left tracheo-bronchial nodes are inoculated separately under the skin of four guinea pigs; forty-five days later, the eight guinea pigs are bearers of the specific adenitis.

GROUP IV — VACCINATED NOS. 7 AND 8 · CONTROLS NOS. 16 AND 16 BIS

The two vaccinated animals are tested, at the same time as the controls 16 and 16 bis, *twelve months after vaccination*. (By reason of this long delay of one year, we thought it necessary to take two controls.) Control 16

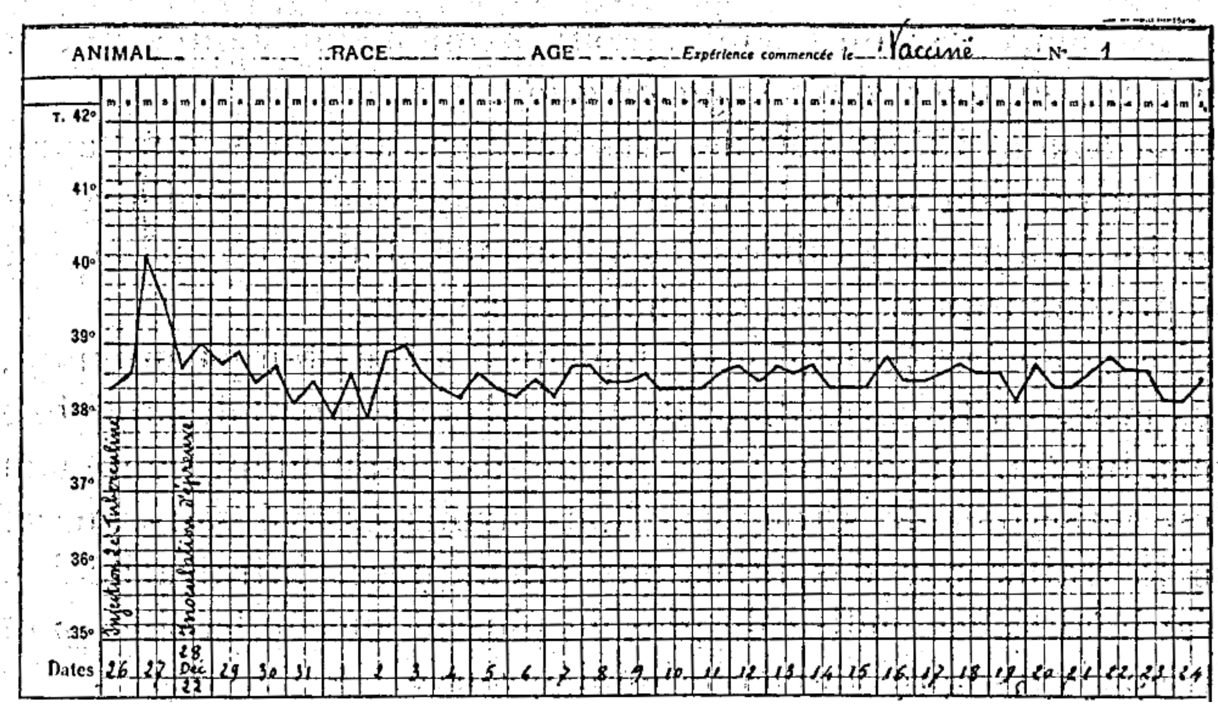

FIG. 6 — Vaccinated no. 1: challenge preceded by tuberculin — the immediate reaction is abolished.

dies thirty-two days after the inoculation: massive pulmonary granulie. Control 16 bis, reduced to the state of a skeleton, dies fifty-eight days after the inoculation: hyperacute miliary tuberculosis. Vaccinated 7 and 8, in excellent condition, are slaughtered three months after the test. They are found free of tuberculosis. Their left pretracheo-bronchial nodes are inoculated separately under the skin of four guinea pigs. Forty-five days later, the eight guinea pigs are bearers of the specific adenitis.

GROUP V — VACCINATED NO. 9 · CONTROL NO. 17

(Vaccinated no. 10 died at the very beginning of the experiment, twenty-five days after vaccination, of ulcerative gastritis.) Vaccinated no. 9 is tested, at the same time as its control 17, *fifteen months after vaccination*. Control 17 dies fifty-four days after the inoculation: acute miliary tuberculosis. Vaccinated no. 9, in excellent condition, is sacrificed for the butcher two months after the test. Its autopsy, made in the presence of the veterinary director of the Lille abattoir, did not permit the detection of any trace of tuberculosis, although its bronchial and mediastinal nodes appeared a little voluminous and succulent on section.

GROUP VI — VACCINATED NOS. 11 AND 12 · CONTROL NO. 18

The two vaccinated animals are tested, at the same time as the control, *eighteen months after vaccination*. Control 18 dies forty-two days after the inoculation: massive pulmonary granulie. After the test inoculation, vaccinated no. 11 makes the immediate thermal reaction we have previously noted, then all returns to order. Vaccinated no. 12 did not make this reaction; barely, in the four days that followed, did its temperature rise a little above normal. On the other hand, from the thirteenth to the eighteenth day, it makes a rather marked hyperthermia (39.8°), after which its health becomes perfect again.

These two vaccinated animals 11 and 12 are, by reason of their excellent condition, slaughtered for the butcher two months after the virulent inoculation. Their autopsy was made at the abattoir in the presence of M. Bossut.

Vaccinated no. 11 is entirely free of tuberculosis. Its bronchial and mediastinal nodes are a little more voluminous and softer than usual.

On opening the thoracic cavity of vaccinated no. 12, one observes that the two lungs are incompletely collapsed. Their consistency is firm. Section shows the pulmonary tissue dense, sclerous, everywhere permeable, but of a somewhat pale tint. One finds in it no tuberculous lesion. The bronchial and mediastinal nodes, doubled in volume, are firm. They contain white, greyish islets surrounded by a bloody, almost haemorrhagic zone, without visible tuberculous follicles.

This last autopsy is interesting on more than one count. One cannot doubt that in this animal the lesions of ganglionic and pulmonary sclerosis are the consequence of the test inoculation. They indicate that this vaccinated animal has, after eighteen months, arrived at the limit of its tolerance towards virulent germs, although its resistance is still sufficient to prevent it from contracting a form of acute granular tuberculosis such as its control shows.

The observations of our last two vaccinated animals attest moreover that, if the persistence of the local vaccinal lesion reflects in some sort the state of tolerance towards virulent bacilli, that state of tolerance manifestly prolongs itself two and three months after the local vaccinal lesion has effaced itself.

Let us return finally to an observation made on two of our animals of groups I and II. We have already said[1] that in animals vaccinated by the intravenous route we recovered the test bacilli, living and virulent, in the bronchial nodes twelve and eighteen months after the inoculation. Now, in our animals vaccinated by the subcutaneous route, we observe that twelve months after the test in one (no. 1), eleven months after in the other (no. 3), the bronchial nodes are no longer virulent.

[1] These *Annales*, February 1913.

What other interpretation could be given of this fact than that massive subcutaneous vaccination confers upon the animals submitted to it a greater aptitude for eliminating the superadded virulent bacilli, and consequently that it is more effective — at the same time as more practical — than vaccinal inoculation by the intravenous route?

IV PRINCIPLES OF A NEW PROPHYLAXIS OF BOVINE TUBERCULOSIS, BASED ON THE FOREGOING EXPERIMENTAL RESULTS

The facts we have reported above demonstrate, better still than those set out in our earlier researches, the possibility of conferring upon young cattle, for a period of time which may extend beyond one year, a manifest resistance to tuberculous infection, even when, from one to fifteen months after vaccination, that infection is realised in the gravest, most constantly and rapidly fatal manner — that is to say by the intravenous route, with a massive dose of highly virulent bacilli.

They bring equally the proof of the perfect innocuity of our vaccine bacillus. This bacillus has lost all aptitude for provoking the formation of tubercles. It is, even at very high doses, harmless for all animal species; it behaves in the organs like a true saprophyte, incapable of recovering its virulence in place. Its eventual elimination, by the milk or by the intestine, offers no danger for man nor for other domestic or wild animals. Its dissemination in the external environment can occasion no damage.

These are precious qualities which none of the human, bovine, equine or avian bacilli possesses — all of them, to varying degrees, virulent and tuberculigenic — whose employment as vaccines has been proposed by certain experimenters (Behring, R. Koch and Schütz, Th. Smith, McFadyean, S. Arloing, H. Vallée), and whose dispersion in nature is perhaps not devoid of drawbacks.

Everyone knows, indeed, that one of the principal reasons which have hitherto caused breeders to set aside the use of these bacilli as "bovo-vaccines" is the fear — moreover quite justified — of spreading tuberculous infection among other animal species, particularly in man (above all in the child), and also in the pig, or again in the small rodent mammals (rats, mice) and in the birds which too often serve as vehicles for the infections of byres and poultry yards.[1]

[1] Normal bacilli, even slightly virulent, killed by heating, by antiseptics (chlorine, iodine, formol, sodium fluoride) or by prolonged immersion in glycerine — whose employment as a vaccine has likewise been attempted — are in general fairly well tolerated. We have been able to establish by numerous experiments, some of them already old (these *Annales*, April 1914), that young cattle injected with such bacilli acquire a weak degree of resistance to virulent test infections, but that this resistance falls away very quickly and cannot be compared with the much more solid and relatively durable resistance which living bacilli realise. The same holds for healthy animals injected preventively, into the venous circulation, with tuberculin at high dose, with bacillary extracts, or with emulsions of bacillary lipoids.

SYNOPTIC TABLE — RESULTS BY ANIMAL (EDITORIAL SUMMARY OF SECTION III, D)

GROUP	ANIMAL	TREATMENT	CHALLENGE INTERVAL	NECROPSY	DESCRIPTIVE PATHOLOGY	GUINEA-PIG ASSAY
I	**1**	BCG 100 mg	1 mo	12 mo · slaughtered	Entirely free of tuberculosis.	0/4 at 45 days
	2	BCG 50 mg	1 mo	5 mo 7 d · died†	No tuberculous lesion; death from verminous bronchitis.	4/4 adenitis at 30 days
	13	Control	1 mo	60 d · sacrificed	Miliary tuberculosis of the lungs, tubercles at all stages.	—
II	**3**	BCG 100 mg	3 mo	11 mo · slaughtered	Free of tuberculosis.	0/4 at 45 days
	4	BCG 50 mg	3 mo	11 mo · slaughtered	Free of tuberculosis.	4/4 adenitis at 45 days
	14	Control	3 mo	44 d · died	Massive pulmonary granulie.	—
III	**5**	BCG 100 mg	6 mo	8 mo · slaughtered	Free of tuberculosis.	4/4 adenitis at 45 days
	6	BCG 50 mg	6 mo	8 mo · slaughtered	Free of tuberculosis.	4/4 adenitis at 45 days
	15	Control	6 mo	60 d · sacrificed	Miliary tuberculosis of both lungs with foci of hepatisation.	—
IV	**7**	BCG 100 mg	12 mo	3 mo · slaughtered	Free of tuberculosis.	4/4 adenitis at 45 days
	8	BCG 50 mg	12 mo	3 mo · slaughtered	Free of tuberculosis.	4/4 adenitis at 45 days
	16	Control	12 mo	32 d · died	Massive pulmonary granulie.	—
	16 bis	Control	12 mo	58 d · died	Hyperacute miliary tuberculosis; reduced to a skeleton.	—
V	**9**	BCG 100 mg	15 mo	2 mo · slaughtered	No trace of tuberculosis; bronchial and mediastinal nodes a little voluminous and succulent on section.	Not reported
	10	BCG 50 mg	*not challenged*	25 d after vaccination · died	Ulcerative gastritis. Vaccinal depot: hard fibrous mass 5 × 2 cm with golden-yellow necrotic islets, no suppuration.	—
	17	Control	15 mo	54 d · died	Acute miliary tuberculosis.	—
VI	**11**	BCG 100 mg	18 mo	2 mo · slaughtered	Free of tuberculosis; bronchial and mediastinal nodes a little more voluminous and softer than usual.	Not reported
	12	BCG 50 mg	18 mo	2 mo · slaughtered	Dense pulmonary sclerosis, everywhere permeable; nodes doubled in volume with white-grey islets — no tuberculous lesion found in them.	Not reported
	18	Control	18 mo	42 d · died	Massive pulmonary granulie.	—

Vaccination: a single dewlap inoculation of BCG in 10 cubic centimetres of physiological water, allocated by number parity — odd numbers 100 milligrammes, even numbers 50. Challenge, for every animal that received one: 5 milligrammes of virulent bovine bacilli, weighed fresh, into the jugular vein. The necropsy column gives the time from challenge (from vaccination for no. 10, which died before its group was challenged). Guinea-pig assays were made on the left pretracheo-bronchial node, four animals per node; "not reported" means the memoir records no assay for that animal, and "—" that none was made. † No. 2 died rather than being slaughtered. Note that the necropsy interval shortens as the challenge interval lengthens — eleven to twelve months in groups I and II against two to three months in groups IV to VI — so time since vaccination is confounded with the time allowed for lesions to appear and for bacilli to be eliminated: the two animals whose nodes proved sterile (nos. 1 and 3) are at once the high-dose animals and the animals given longest to clear. This table does not appear in the original memoir; it collects the narrative of section III, D.

With our BCG bacillus nothing of the sort is to be feared. We have even excellent reasons to believe (and we shall return later to this subject of capital importance) that its wide use will contribute to rarefying tuberculosis-the-disease, by generalising, above all in young subjects, latent infection by a living, avirulent and non-tuberculigenic bacillus.

For ourselves, indeed — and we have affirmed and demonstrated it since 1906 — antituberculous immunity (or, if one prefers, resistance to reinfections) is linked to the presence of a few bacilli, few in number and of low virulence, in the organism. So long as the latter remains thus parasitised, without that parasitism entailing serious cellular or functional disorders, reinfections — provided they be neither too frequent nor massive — will have upon it no other effect than to increase its sensitisation (or its intolerance) towards bacillary microbial bodies of whatever kind, living or dead, and towards tuberculin.

But this immunity ceases as soon as the initial "vaccinating" symbiosis itself ceases to exist, whether because the vaccinating bacilli have been destroyed by the normal processes of cellular digestion, or because they have been eliminated by the natural emunctories of microbes (bile, intestine, mammary glands). And then the virulent bacilli of reinfection — if any already exist which have remained intact, or if new ones have been introduced into an organism whose immunity is thus extinguished — resume or retain their full value as pathogenic bacilli, such as they would have towards an organism virgin of any earlier infection.

Is this then a true immunity, in the sense bacteriologists give to that word, that is to say a state refractory to the disease?

We consider that this is not deniable, for this immunity is perfectly analogous to that realised by living attenuated virus-vaccines, such as the Jennerian vaccine, the anthrax vaccine, that of swine erysipelas or that of rabies. It is, like the immunity these virus-vaccines determine, more or less durable according to the nature, the quantity and the virulence of the germs which produce it.

It would be strange if one demanded of an antituberculous vaccine anything other than what one demands of an anti-anthrax vaccine, for example — that is to say, to be effective and harmless. Everyone knows that the immunity acquired by vaccination against anthrax or against swine erysipelas hardly lasts more than a year, and that the immunity acquired by Jennerian vaccination or by antirabic vaccination fades away little by little in the space of about seven years. Why should it have to be otherwise for artificial immunisation against tuberculosis?

It is only true that the latter comes up against a difficulty which the others do not know: it is that it is difficult to apply, at least in the present state of most livestock holdings, to adult subjects, because these are too often and to varying degrees already tuberculised, or infected by a few virulent bacilli. In such subjects the injection of vaccine bacilli — as, moreover, that of other bacilli, virulent or attenuated, or even dead —

determines an increase of sensitivity to tuberculin which is made manifest by the appearance of the "phenomenon of Koch".

It is therefore preferable, for this reason, to institute methodically vaccination against tuberculosis by addressing oneself to the very young subjects, from the first days following their birth, before they have been exposed to occasions of infection — abundant, or even discreet but repeated — by virulent bacilli.

Such are the considerations which, supported by our experiments, decided us to accept the generous proposals made to us to try the preventive effects of our vaccine bacillus BCG in agricultural holdings long gravely infected with tuberculosis.

From the first days of 1921, M. Lallemant, prefect of the Seine-Inférieure, having been informed of our researches following a lecture by one of us, was good enough to offer us access to one of the farms dependent upon his administration. Almost immediately afterwards, M. Le Grand, conseiller général of the same department, placed at our disposal his important holding of Gruville.

With the concurrence of the departmental agricultural office, and the precious help of M. Richart, veterinary director of the sanitary service, and of M. Marcel Boissière, veterinary surgeon at Valmont, we were able to undertake an important series of trials which we count on pursuing for several consecutive years. Experiments of the same nature were begun more recently (January 1922) in Seine-et-Oise, thanks to the obligingness of M. Pierre de Chezelles and the concurrence of M. Brinet, veterinary surgeon at Magny-en-Vexin. To all these kind collaborators we address our most grateful thanks.

The problem we proposed to ourselves to resolve was the following:

> **In a holding infected with tuberculosis, without changing anything whatever in the mode of existence or the housing of the animals, without modifying the usual methods of rearing the young, is it possible, by the normal play of births, by vaccinating the newborn in the first fifteen days of their life and revaccinating them each year, to purge that holding of tuberculosis[1] within a period of five years?**

We evidently cannot, as of now, speak of results, by reason of the very form of the question posed. But it is not without interest to make known the course of our operations.

At the beginning of these operations, in January and in the course of the year 1921, we had thought — knowing nothing of the trials carried out by H. Vallée in a somewhat different sense (but proceeding from the same idea), which he has recently published[2] — of creating under the skin of the animals a persistent and active

[1] By tuberculosis we understand tuberculosis-the-disease, contagious and progressive, entailing in any manner whatever economic losses for the breeder. We naturally exclude latent infection, not manifested by follicular lesions or by tubercles.

[2] *Académie des Sciences*, 2 January 1921.

local lesion. To that end, 28 calves born in the two farms of the Seine-Inférieure underwent, within the first month of their birth, the insertion into the subcutaneous connective tissue of the dewlap of a ball of pumice stone of 10 millimetres, charged under vacuum with a concentrated emulsion of vaccine bacilli.

Parallel trials, carried out in the laboratory, allowed us to follow the destiny of the bacilli thus immobilised and protected. We were able to establish that they degrade rapidly, and that the resistance of the calves so treated to the virulent experimental test disappears very quickly, as does their aptitude for reacting to tuberculin.

We therefore abandoned this method and, so as not to make our 28 calves lose the benefit of the partial immunity which the ball might have conferred upon them, we vaccinated them all in 1922, at the same time as 32 other fresh, newborn animals, by intravenous inoculation of 20 milligrammes of BCG, a dose whose efficacy we knew from our experiments of 1920.[1]

From January 1923, better enlightened by the preliminary experiments which suggested to us those forming the principal object of this memoir, we have employed exclusively the method of preventive inoculation by the subcutaneous route. This has up to the present been put into practice for the annual revaccination of 58 animals born in 1921 and 1922, and for the first vaccination of 48 young animals born in the course of the year 1923.

All that we can at present affirm is that *this intervention is absolutely harmless*. At no moment have the animals which underwent it in the first fifteen days of their birth experienced the least malaise. They have all remained in perfect health.

Here, moreover, is the proof that revaccinations made at twelve months' interval present no drawback.

> EXPERIMENT. — Four heifers vaccinated a first time are revaccinated one year later. In three of them the traces of the initial injection in the dewlap are still perceptible. The local reaction observed differs in nothing from that which was observed at the time of the first inoculation. None of the animals made an immediate or a late thermal reaction.

In February 1924, in the holdings where we operate, 142 animals of one, two and three years had been thus revaccinated without incident of any kind.

The reader perhaps asks himself why we have adopted annual revaccination, when the resistance conferred by a first inoculation is prolonged, according to our experiments, at least until eighteen months. To this we answer that, the operation being harmless, we believe it preferable not to wait for the limit of its efficacy.

To this reason one may add an argument of a psychological order: since Pasteur, those who devote themselves to livestock rearing know that they can, by an annual vaccination, preserve their beasts from

[1] These *Annales*, September 1920.

some of the grave maladies which formerly decimated them. It is therefore desirable, for the prevention of tuberculosis, to keep that term as an established habit, and thus to bring no disturbance to the economic life of agricultural holdings.

Subject to the complementary lessons which may be brought to us by the extension of our trials of vaccination of young cattle against tuberculosis in the farms of Seine-Inférieure and Seine-et-Oise which have offered to help us in our researches, we believe ourselves founded, as of now, in extending these experiments to the rearing centres which have technical means sufficient for them to be rigorously supervised and followed.

It is important, indeed, that breeders be put as soon as possible in a position to be protected against bovine tuberculous infection by means more effective than those, as draconian as they are inoperative, which the present legislation on sanitary policing places at their disposal.

The latter prescribes, it is true, the isolation of tuberculous animals, or even of animals merely reacting positively to tuberculin, but it does not provide the means of rendering that isolation complete and real. It does indeed order the disinfection of contaminated byres, but it is silent upon the measures to be applied when that disinfection is practically impossible, by reason of the defectiveness or the decrepitude of the premises.

Doubtless the efforts attempted since the brilliant campaigns of Bang in Denmark and of Nocard in France, with a view to the prophylaxis of bovine tuberculosis by the tuberculination of livestock, have produced some useful effects. But the disappointments of the veterinary surgeons who have applied the "Bang method" systematically and with the greatest care have become numerous in all countries, although this method has certainly contributed to reducing tuberculous morbidity notably. It has happened many times that animals recognised as free of infection by the absence of reaction to tuberculin, but coming from contaminated byres and introduced into environments where all interior contagion was avoided, have nevertheless manifested, after variable times and often after a calving, a positive tuberculin reaction.

Such so-called "failures of the Bang method" are easily explicable today, now that we know with what slowness a discreet bacillary infection sometimes creates the symbiotic cellular lesion which alone renders the organism that bears it sensitive to tuberculin. In cattle, as in man, *the tuberculin reaction may become positive only long months, sometimes several years, after the contamination has been realised.*

One cannot therefore count upon tuberculin to set aside suspect subjects. And, in these conditions, it appears assuredly preferable to vaccinate the whole herd, beginning with the very young animals, which are much more sensitive than adults to bacillary infection, and upon which, as we have established, it is possible to confer a manifest and sufficiently durable resistance to grave contaminations, even artificially provoked.

CONCLUSIONS

1. By addressing oneself to the very young animals of the bovine species, born less than two weeks before, and by injecting under their skin, in the full loose connective tissue of the dewlap, a strong dose (50 to 100 milligrammes) of our bacillus modified by a long series of successive cultures upon potato cooked in ox bile glycerinated at 5 per 100 (BCG bacillus), one can vaccinate them against virulent bacillary infection, to the point that they do not contract tuberculosis if one injects into their veins, up to the fifteenth month after vaccination, a dose of living, virulent bovine bacilli capable of killing by acute granulie all the controls of the same age in less than two months.

2. Young cattle thus vaccinated react positively to tuberculin during the whole time that their immunity is establishing itself. Some months after they cease to react, this immunity tends to disappear, and the moment arrives when they behave like unvaccinated animals.

3. Vaccination with our BCG bacillus is harmless, not only for young cattle and for adult cattle free of tuberculosis, but also for all animal species susceptible of being infected by the tubercle bacillus.

4. This BCG bacillus is completely deprived of virulence. It has lost all aptitude for provoking the formation of tubercles. It nevertheless remains toxic for the tuberculous animal. The glycerinated broths in which it is cultivated contain tuberculin, and the organisms into which it is injected produce antibodies detectable by the Bordet-Gengou reaction.

5. We consider it desirable that wider trials of prophylaxis against tuberculosis of livestock, by the method we have studied, be made in some large agricultural holdings where an effective sanitary supervision could be regularly exercised. We are entirely disposed to facilitate in this sense the experiments that may be undertaken.

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